

Smart Email Prioritization with Adaptive Learning

CS6140 Final Project Proposal

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Problem Description

Email overload is a widespread problem — users struggle to distinguish important messages from newsletters, notifications, and spam. Existing filters apply fixed rules with **no personalization**. We propose a system that classifies emails by priority and **continuously learns from real-time user feedback**, becoming more accurate for each individual user over time.

Data Summary

Our primary dataset is the Enron Email Dataset [1], available on Kaggle, which contains over 500,000 real emails from 150 employees of the Enron Corporation. This dataset offers rich diversity in communication styles, topics, and email structures, making it well-suited for training a multi-class email prioritization model. **User feedback collected through the application will serve as additional personalized training data**, enabling the model to adapt to individual preferences over time.

Methods

We will apply standard text classification techniques — including TF-IDF feature extraction from email subject lines and body text — and train machine learning models to categorize emails into priority tiers [2]. **A real-time feedback mechanism will allow the model to incrementally update based on user corrections**, enabling deep personalization over time, similar to how modern recommendation systems adapt to user behavior.

Two key challenges are anticipated. First, **class imbalance** between high- and low-priority emails may bias the model toward the majority class; we plan to address this through resampling techniques or weighted loss functions. Second, the **cold start problem** means new users have no feedback history to draw from, requiring a well-tuned general-purpose baseline model before meaningful personalization can take effect.

Preliminary Results

We have confirmed access to the Enron Email Dataset on Kaggle, reviewed its structure and label distribution, and completed an initial exploration of email content and metadata fields. Development environment and project responsibilities have been divided among team members.

References

- [1] Klimt, B. and Yang, Y. The Enron Corpus: A New Dataset for Email Classification Research. European Conference on Machine Learning (ECML), 2004.
- [2] Alsuwit, M. H., Haq, M. A., and Aleisa, M. A. Advancing Email Spam Classification using Machine Learning and Deep Learning Techniques. Engineering, Technology & Applied Science Research, 14(4), 14994–15001, 2024.